



LECTURE 1: INTRODUCTORY CONCEPTS

Introduction

- ▶ Clinical pathology is a “subspecialty of pathology that deals with the use of laboratory methods for the diagnosis and treatment of disease
- ▶ In general terms, it is the study of disease in the clinical environment by use of laboratory assays
- ▶ Laboratory tests should be used with other diagnostic procedures; two diagnostic procedures are imperative: (1) obtain a complete history, and (2) perform a complete physical examination. With knowledge gained from these two basic procedures, a diagnostician can select diagnostic procedures to clarify or classify identified problems.

Importance of analyzing samples via laboratory procedures



- ▶ To detect an unidentified pathologic state
- ▶ To define, classify, or confirm a pathophysiologic disorder or disease state
- ▶ To eliminate (rule out) a possible cause of the animal's illness
- ▶ To assess changes in a pathologic state either due to natural progression of the disease or because of medical or surgical therapy

SAMPLES

- ▶ Most clinical laboratory assays are designed to detect or quantify substances or cells in blood samples; the substance or cell of interest is called the analyte
- ▶ Obtaining useful results for the analyte requires appropriate samples
- ▶ Whenever there is doubt about the appropriate sample for a particular test at a particular laboratory, the laboratory should be contacted prior to sample collection

SAMPLES (Blood)

- ▶ Blood and its major components are frequently used as samples for laboratory assays; must be collected and processed properly
- ▶ composed of blood cells and plasma
- ▶ immediately mixed with an anticoagulant to prevent initiation of clot formation and to maintain cells and other components in suspension
- ▶ Analysis or processing of whole blood must be relatively rapid because the cells die within a few hours; unacceptable for analysis
- ▶ samples must be analyzed within minutes, usually within hours rarely within days

SAMPLES (Plasma)

- ▶ fluid component of blood that is harvested after centrifugation of an anti coagulated blood sample
- ▶ contain the anticoagulant that can interfere with some assays

Anticoagulants used for blood sample collection

a. Calcium-binding agents prevent Ca_{2+} from participating in the formation of a blood clot/

(1) EDTA (as Na_2EDTA , K_2EDTA , or K_3EDTA)

- ▶ preferred anticoagulant for almost all routine hematologic tests, including the complete blood count (CBC) assays

SAMPLES

(2) Citrate (as sodium citrate)

(a) preferred anticoagulant for most tests of the coagulation system. Ca_{2+} is added to the citrated plasma to override the effects of citrate and enable coagulation enzymes to function. Citrate's anticoagulant activity is achieved by its forming an ionic bond with Ca_{2+} .

(b) Because it has low toxicity, citrate is also preferred for collection of whole blood to be used for transfusions.

SAMPLES

(3) Oxalates (as lithium, ammonium, and potassium salts)

(a) few laboratory tests; for example, oxalate is the anticoagulant in sodium fluoride tubes that are used for glucose and lactate assays; distort morphologic features of leukocytes and erythrocytes and thus are unsuitable for hematologic samples

(b) Oxalate's anticoagulant activity is achieved by its forming an ionic bond with Ca_{2+}

SAMPLES

Heparin (as lithium, ammonium, potassium, or sodium salts) ; inhibits coagulation factors (including thrombin)

(1) Used for several special laboratory assays (such as blood gas analysis) and can be used for many clinical chemistry assays

(2) Major disadvantages

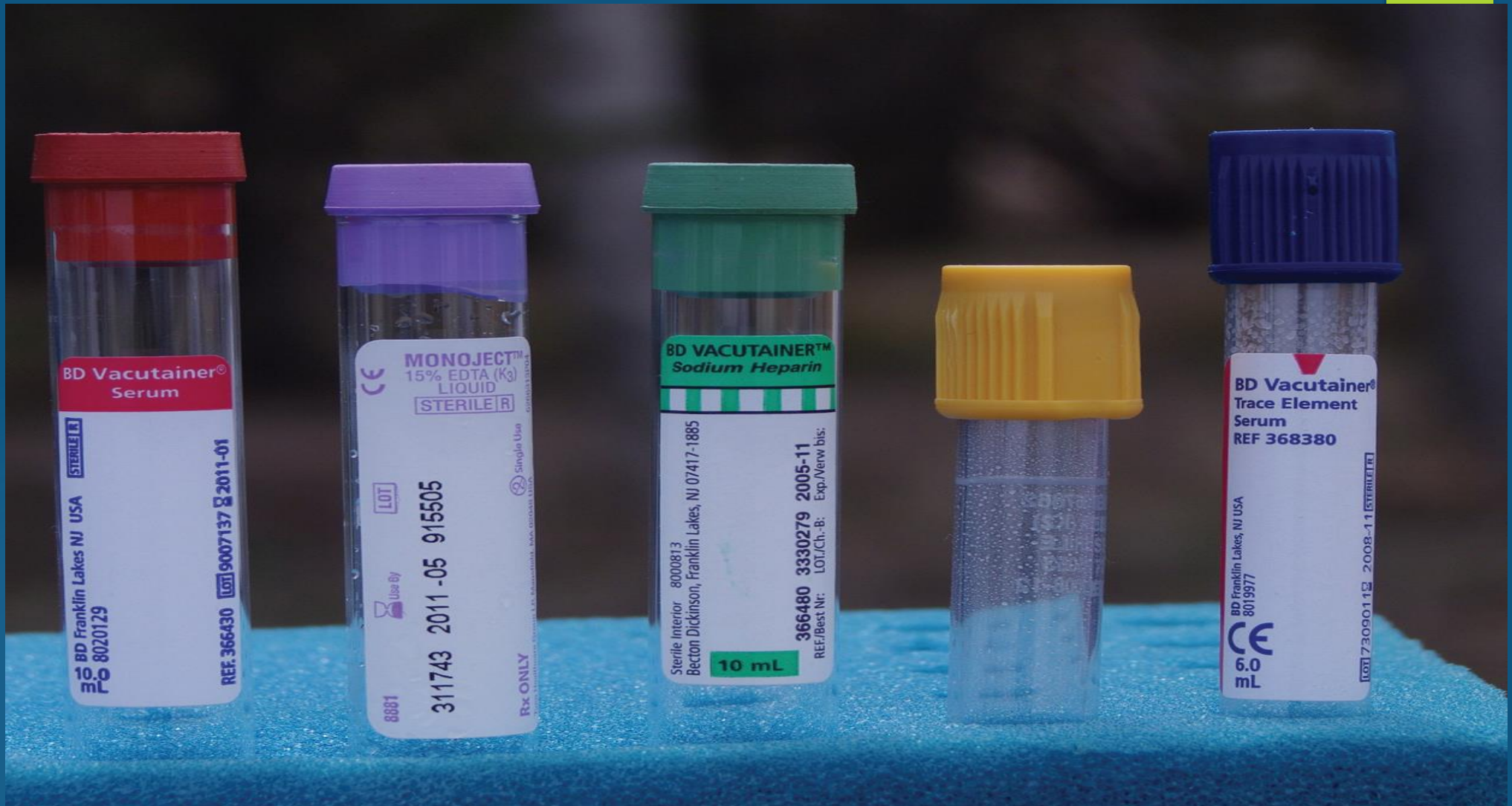
(a) Alters morphologic features and staining of leukocytes

(b) Allows clotting as effects are slowly overridden by the coagulation system

(c) Allows platelet clumps to form

SAMPLES (serum)

- Serum is the fluid component of blood that is harvested after centrifugation of a coagulated (clotted) blood sample
- To get the maximal amount of serum from the clotted sample, centrifugation should not be started prior to the retraction of the clot (which typically takes at least 30 min if a clot activator is not present in the tube)
- If samples are centrifuged prior to clot retraction, some serum will be trapped in a soft fibrin clot
- same composition as plasma except serum does not contain most of the coagulation major protein (on a weight/volume basis) that is absent in serum but present in plasma is fibrinogen



Examples of commercial blood collection vials. From the left, the red stopper indicates no anticoagulant (“clot tube”), the lavender one (often called “purple top”) has EDTA, the green has heparin, and the yellow and blue plastic capped tubes are specialized tubes for specific tests.

Tube type	Color code	Additive	Application
NO ADDITIVE TUBE		-	Immunology, serology, biochemical test
COAGULANT TUBE		Clot activator	
COAGULANT TUBE WITH GEL		Clot activator, Separating gel	
HEPARIN TUBE		Sodium Heparin	Biochemical tests
		Lithium Heparin	
		Lithium Heparin, Separating gel	
		Sodium Heparin, Separating gel	
PT TUBE		Sodium Citrate (1:9) 0.109mol/l or 0.129mol/l	Coagulation studies
GLUCOSE/LACTATE TUBE		Potassium oxalate, Sodium fluoride	Blood sugar examination
		EDTA, Sodium fluoride	
EDTA TUBE		EDTA K2	Blood routine examination, molecular diagnostics
		EDTA K3	
		EDTA K2, Separating gel	
ESR TUBE		Sodium Citrate (1:4) 0.109mol/l or 0.129mol/l	Blood cell sedimentation rate test

Handling and storage

- ▶ Separation: For hemostasis and chemistry testing, the plasma or serum should be separated from cells as soon as possible after sample collection by centrifugation
- ▶ Labeling: All body fluid samples taken from a patient should be correctly labeled with the patient name or identification and the type of specimen (e.g. serum, plasma, synovial fluid, peritoneal fluid)
- ▶ Storage: All fluid samples should be stored at 4°C until submission. In contrast, slides should not be refrigerated (the cells lyse with storage)

Submission

- ▶ Label properly: Make sure all tubes and slides are labeled with patient identification and the contents (e.g. blood, urine)
- ▶ Package appropriately: Make sure the samples are protected from breakage. Cardboard slide boxes frequently break during transit, so if you are using them, wrapping them in bubble wrap or other cushioning material will protect them
- ▶ Add cool packs: Ship fluid samples on cool packs. The samples should not be in direct contact with the cool pack, but wrapped in paper towels. Direct contact of cells with an ice pack will cause freezing of the sample and cell lysis

Submission

- ▶ Avoid formalin: Do not ship formalin containers in the same package as the slides or tubes. Formalin readily leaks out of containers and affects the quality of the samples
- ▶ Ship ASAP: The quicker the sample gets to the laboratory, the fewer the false changes in results due to storage
- ▶ Provide a good history: This includes signalment (species, age, breed, sex of patient), history that may be relevant (e.g. travel, access to toxins, current medications), pertinent clinical signs (e.g. epistaxis) and, for cytology, a good description of the aspirated lesions (including imaging findings), e.g. multiple hypoechoic masses in the liver

Packaging diagnostic samples

Below are the video link for :

- General packaging

<https://www.youtube.com/watch?v=2byIL4EFha8>

- packaging tips: blood samples, fresh and fixed tissue, and cytology slides and answer the following questions:

<https://www.youtube.com/watch?v=JwcsQcQshBs&t=133s>

Major Types of Laboratory Assays

A .Clinical hematology assays

1. Quantitation of cell concentrations in blood
2. Semiquantitation of cell concentrations
3. Defining or classifying cells by microscopic features
4. Assessing the coagulation properties of blood

B. Clinical chemistry assays

1. Detecting or quantifying the concentration of a chemical substance
 - A.Quantitative analysis
 - B.Semiquantitative analysis
 - C.Qualitative analysis
- 2.Detecting or quantifying the activity of a chemical substance
 - A.Quantitative analysis
 - B.Qualitative analysis

C. Clinical microscopy

1. Clinical cytology
2. Surgical histopathology
3. Urine sediment analysis
4. Clinical parasitology



Basic Laboratory Safety

- Sharps
- Chemicals
- Biological materials
- PPE (personal protective equipment)